Unveiling Bias: Visualizing Gender Bias in Word Embeddings

In recent years, artificial intelligence (AI) has been used widely in decision-making areas like hiring, education, healthcare, and law enforcement. While these systems are said to be objective, research has shown that they often carry over biases from the data they're trained on (Sankaranarayanan, 2024). This means they can unintentionally reinforce harmful stereotypes and make existing social inequalities worse.

In this essay, I'm focusing on bias in AI through something called the Word Embedding Association Test (WEAT). WEAT is a method used to measure biases in word representations that AI systems learn. It's a way to uncover how AI picks up on and encodes systemic inequalities from the language it processes. Beyond just looking at text, I also explored how multimodal AI systems, like DALL-E, reproduce stereotypes in visual outputs. Through this, I wanted to see how useful visualization techniques are in spotting these patterns. Ultimately, AI needs to be reimagined to prioritize fairness and inclusivity—not just through better technology, but also by reflecting on the ethics behind it. By analyzing word vectors, my goal is to add to the conversation about how we can create more transparent and accountable AI systems.

The Word Embedding Association Test (WEAT)

WEAT was introduced in 2017 by Aylin Caliskan, Joanna J. Bryson, and Arvind Narayanan. It's based on ideas from psychology, specifically the Implicit Association Test (IAT), which measures human biases (Caliskan et al., 2017). But instead of analyzing people, WEAT looks at how AI associates words. For example, does AI connect "man" more strongly with "engineer" and "woman" with "nurse"? This test can reveal gender stereotypes embedded in AI. Sabbaghi and Caliskan's work showed me how AI often encodes societal stereotypes in its word embeddings.

Their work inspired me to think critically about the hidden associations in language and how they can be uncovered. One of the key insights from their research is that these biases are often not immediately visible. AI models absorb patterns from the data they are trained on, which means unintended associations can remain hidden without careful analysis. This motivated me to explore how visualization techniques could make these hidden patterns more transparent and accessible.

Data Biography

For this project, I used GloVe (Global Vectors for Word Representation), a dataset created by researchers at Stanford in 2014. GloVe was trained on vast amounts of data, like books, Wikipedia arcticles, and web content (Pennington et al., 2014). The goal was to train AI to understand the relationships between words. For example, if "doctor" appears near "hospital", GloVe learns and represents those words closer. This allows AI to understand spoken language based on patterns in text (Qadir, 2023).

GloVe is widely used in AI systems, including tools like ChatGPT (Hafiz & Aisyah, 2023). But since GloVe was trained on internet data, books, and encyclopedias, it also reflects the biases and stereotypes found in those sources.

Limitations

While WEAT and GloVe are powerful, they have their limitations. For example, GloVe denotes words as set vectors; it cannot adjust for context. It sees the word “bank” as the same whether it’s referring to money or a riverbank (Kondurkar et al., 2024). This oversimplification can lead to decisive results

Generating the Visualizations

To explore gender biases, I replicated and adapted WEAT’s methodology using natural language processing (NLP) techniques in Python to analyze word embeddings from the GloVe model. By calculating how gendered terms relate to attributes like professions, traits, and activities, I was able to create visualizations that show these associations. These visualizations allowed me to see the patterns more clearly and are a key part of my analysis.

Visualizations

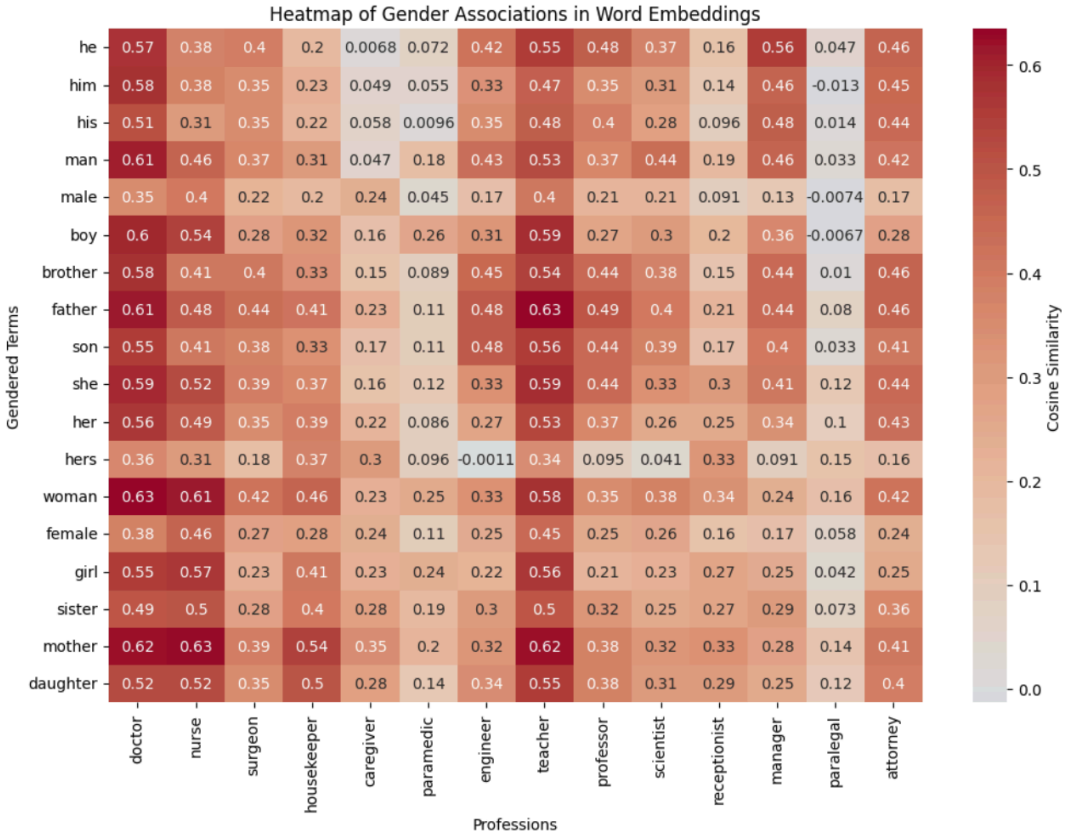


Figure 1: Heatmap of Gender Associations in Word Embeddings
Cosine similarity scores between gendered terms and professions in GloVe embeddings, highlighting gendered patterns.

This heatmap (see *Figure 1*) visualizes the cosine similarity between gendered terms (e.g., “he,” “she”) and profession-related terms (e.g., “doctor,” “nurse”) within the GloVe word embeddings. Each cell represents the degree of association between a gendered term and a profession, with darker shades indicating stronger associations. The results show some

pretty clear gendered patterns. For example, words like “woman” and “mother” are more closely associated with roles like “nurse” and “housekeeper,” while “man” and “father” are linked to “doctor” and “engineer.” This highlights how word embeddings reflect societal stereotypes—female-related terms often connect to caregiving and service roles, while male-related terms are tied to leadership and technical fields.

The heat map was used for this purpose since it was best suited to clearly and simultaneously display complex sets of relationships between variables such as gendered terms and professions. This color gradient intuitively shows cosine similarity values that can more easily be communicated to audiences unfamiliar even with NLP. The grid layout facilitates clear comparisons across rows and columns, which makes the emergence of trends, like the consistency of female terms with caregiver roles, more readily apparent. This is a direct application of Edward Tufte's principle of maximizing data density within readability, supporting both high-level overviews and detailed analyses (Tufte, 2001).

This may also indicate that "she" is relatively strongly associated with traditionally male professions, such as "professor" or "scientist," compared with other female terms, such as "her." That would suggest a certain nuance in the ways that gendered pronouns reflect shifting stereotypes—even within biased embeddings. Visualization of such associations supports the essay's claim that AI systems will perpetuate systemic inequities unless the foundational models are examined for bias and those biases are addressed.

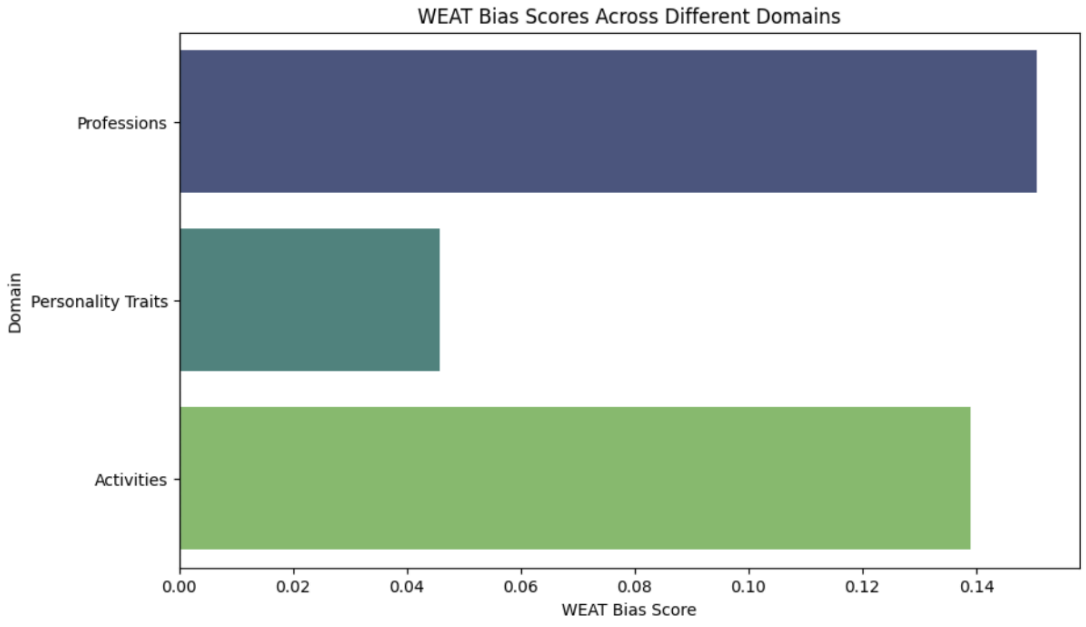


Figure 2: Bar Chart of Domain-Specific Biases
Comparison of average WEAT bias scores across professions, traits, and activities, professions showing the strongest bias.

Figure 2 summarizes the mean WEAT bias scores across the profession, personality traits, and activities in three domains of word embeddings, showing the extent of encoded gender bias: 0.14 for the professions domain, indicating the most powerful gendered associations, followed by activities, which scored 0.12, and personality traits with 0.08. These findings suggest that AI systems trained on word embeddings are more likely to reflect societal

stereotypes in professional settings, such as associating STEM roles with men and caregiving activities with women. This would mean that professions, due to their historical and cultural gender divide, remain highly susceptible to gender bias in AI systems.

The bar chart was chosen for its simplicity and ability to provide a clear, comparative overview of bias across domains. By representing domains on the y-axis and bias scores on the x-axis, the chart emphasizes relative differences, making it easy for the audience to identify which areas require the most attention. This visualization adheres to Colin Ware's principle of perceptual effectiveness, as bar lengths provide a visually intuitive representation of magnitude, ensuring accessibility for both technical and non-technical audiences (Ware, 2010).

A key insight is the unexpectedly high bias score in activities, suggesting that everyday actions like "cooking" or "driving" are also deeply gendered in language representations. It also provides a broader perspective on how biases vary by context, illustrating that some domains (like professions) are more prone to entrenched stereotypes than others. This supports the essay's argument that biases in AI extend beyond professions to more subtle aspects of human behavior. The visualization shows how important it is to remedy systemic inequities perpetuated by AI, mainly in domains that shape decisions and perceptions in society.

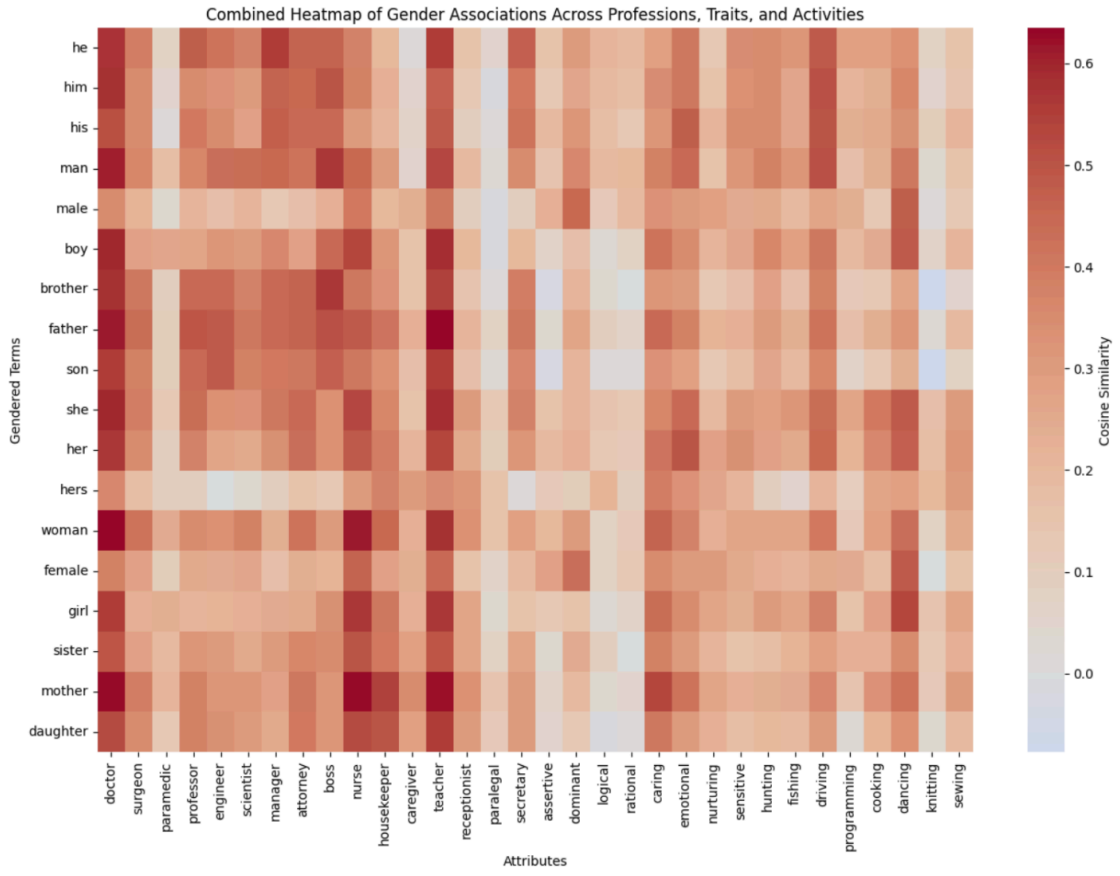


Figure 3: Combined Heatmap of Gender Associations Across Domains

Cosine similarities between gendered terms and attributes across professions, traits, and activities, showing systemic biases.

To provide a holistic view of gender biases across multiple domains, a combined heatmap was created (see *Figure 3*). This visualization combined associations between gendered terms—for example, "he" and "she"—with attributes from three domains: professions, personality traits, and activities. Combining the three domains into one figure, the combined heatmap represents a comprehensive snapshot of the way biases manifest across a variety of contexts.

This combined heatmap was selected because it shows patterns across diverse attributes in one unified visualization. Mapping associations in a grid format allows for cross-domain comparisons without loss of clarity. The color gradient intuitively leads the audience to areas of high bias, making this visualization adhere to Tufte's principle of high data-to-ink ratio—it is dense in information yet not overwhelming. By putting professions, traits, and activities in contrast, the heat map supports multi-dimensional analysis, important to uncover how biases come out along different dimensions of language.

What stands out in this combined heatmap is the systemic nature of these biases. The associations are not confined to a single domain but appear consistently across professions, traits, and activities. However, an unexpected insight is the subtle but notable overlap between female-associated terms and certain neutral traits, such as “logical” and “rational.” This suggests that while traditional stereotypes dominate, word embeddings may also encode traces of shifting societal perceptions.

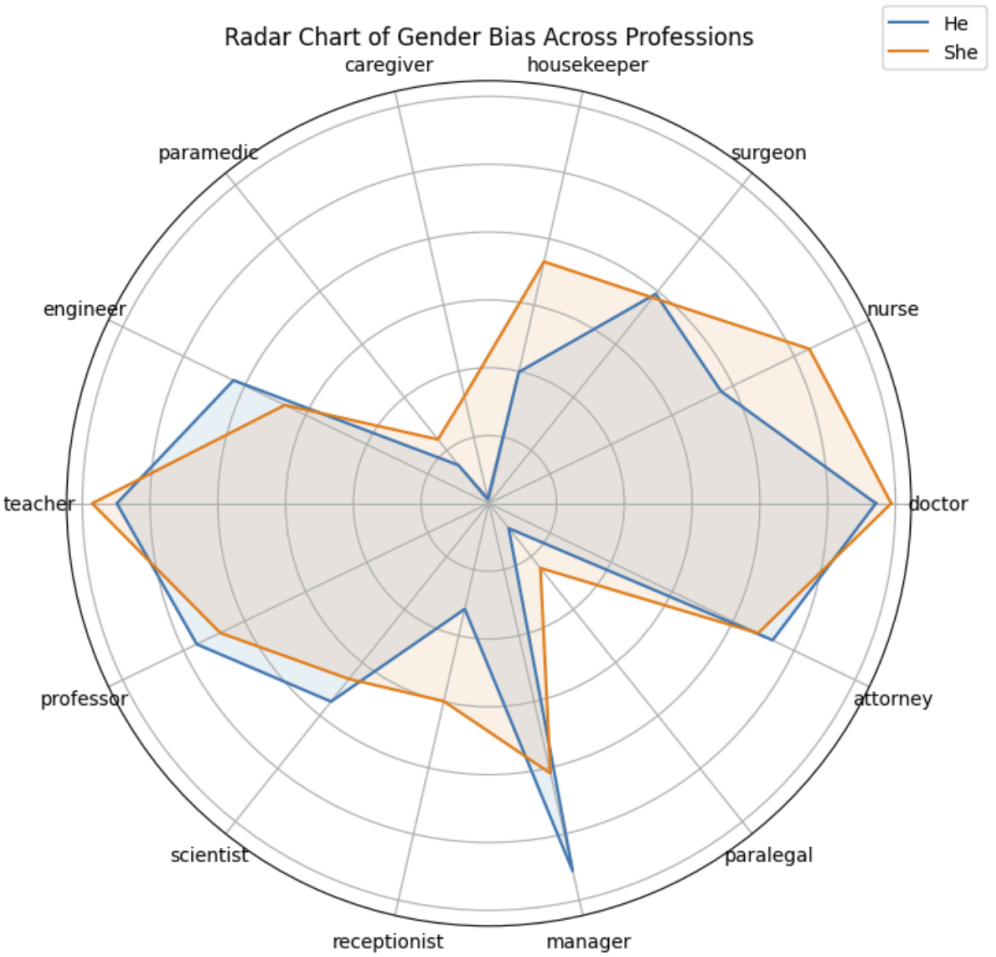


Figure 4: Radar Chart of Gender Bias in Professions

Cosine similarity scores between "he," "she," and various professions, illustrating disparities in gendered associations.

Figure 4 provides a comprehensive view of how male and female terms align with various professions, putting the emphasis on gender biases encoded in word embeddings. Complementing the above heatmap, the radar chart below offers a different insight into word alignment, as it allows immediate comparison across multiple roles in one view.

Each axis here represents a profession, and the values give the cosine similarity between the gendered terms and that profession. Notably, the overlap in professions like "teacher" suggests areas of less pronounced bias, maybe reflecting changes in societal attitudes.

The radar chart is effective in showing multi-variable comparisons across a circular layout and offers a unique view of bias distribution. Its design is particularly fit for capturing the disparity between two categories—male and female—across multiple attributes, or professions. This aligns with principles of effective data visualization, such as ensuring clarity and enabling pattern recognition, as advocated by Tamara Munzner's framework for visualization analysis (Munzner, 2014). The use of a shared scale and contrasting colors (blue for "he" and orange for "she") enhances accessibility and interpretability for diverse audiences.

An interesting insight is the dramatic divergence for roles like “engineer” and “nurse,” where the bias toward one gender is stark, compared to professions like “attorney,” which exhibit more balanced associations. This visualization reinforces the essay’s argument that biases in word embeddings perpetuate entrenched stereotypes about gender roles in the workplace. By highlighting these disparities, the radar chart serves as a compelling tool for advocating reforms in AI training to foster equitable representations of gender across professions.

Exploring Visual Bias in AI

Building on the textual analysis of gender biases in word embeddings, this section explores how such biases extend into visual domains through the lens of DALL-E, a generative AI model designed to create images from textual prompts (OpenAI, 2021). By examining DALL-E’s outputs for various professions, this investigation aims to uncover whether the stereotypes present in language data are reflected in AI-generated imagery.

Methodology and Approach

DALL-E was prompted with simple, profession-based terms—“generate the image of a …”—to analyze the visual representations associated with these roles. The intent was to observe patterns in gender depiction, attire, and settings without imposing additional descriptors that might bias the results further. This approach aims to uncover default associations generated by the model.

Findings

The images produced by DALL-E reveal strikingly gendered patterns (see *Figure 5*). Managers were predominantly depicted as male, characterized by formal attire such as suits and ties, often situated in corporate office environments. Their poses and expressions conveyed authority, decisiveness, and professionalism. Scientists, engineers, and professors were also depicted as male, seemingly connecting high-level knowledge with masculine traits. In contrast, nurses, caretakers, etc were overwhelmingly represented as female. Their expressions and body language emphasized empathy, warmth, and nurturing qualities.

Notably, the absence of explicit gender descriptors in the prompts highlights the influence of underlying training data on these representations. DALL-E’s reliance on vast datasets from the internet and other text-image sources has led it to internalize cultural norms and stereotypes (Caliskan, 2023). These visual biases align closely with the textual associations identified in word embeddings, where male terms are strongly linked to leadership and technical roles, and female terms to caregiving and supportive professions.

generate the image of a surgeon




Here are the generated images of a skilled surgeon in a modern operating room. Let me know if you'd like any adjustments or additional details!

generate the image of a nurse




Here are the images of the nurse as described. Let me know if you'd like any adjustments or additional details!

generate the image of a manager



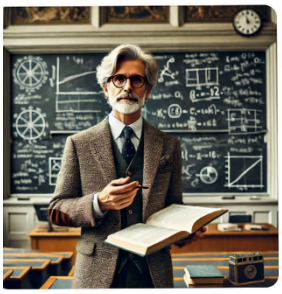
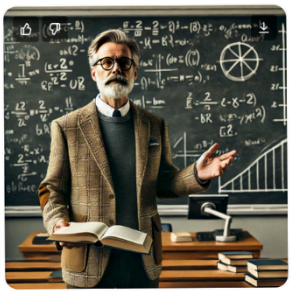

ere are the generated images of a confident manager in a modern office setting. Let me know if ere's anything you'd like adjusted or customized!

generate the image of a receptionist




Here are the generated images of a friendly receptionist in a modern office environment. Let me know if there's anything specific you'd like to adjust!

generate the image of a Professor

lere are two images of a distinguished professor delivering a lecture in a university setting. Let me now if you'd like any refinements or additional elements!

generate the image of a Primary school teacher




Here are two images of a cheerful primary school teacher interacting with young students in a vibrant classroom. Let me know if you'd like any modifications!

generate the image of an Attorney




are two images of a confident attorney presenting in a courtroom. Let me know if you'd like adjustments or additional elements!

generate the image of a Paralegal




Here are two images of a professional paralegal working in a modern law office. Let me know if you'd like any changes or additional details!

generate the image of an engineer



Here are the images depicting a professional engineer in a modern high-tech environment. Let me know if you'd like to adjust or add any specific details!

generate the image of a Housekeeper



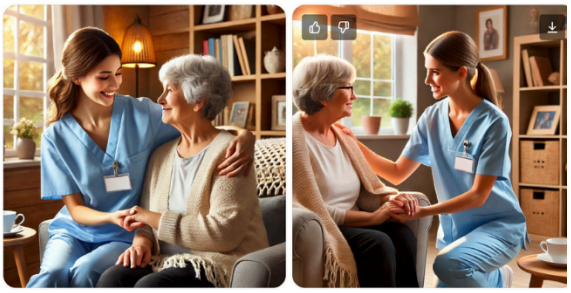
Here are two images of a professional housekeeper in a clean and welcoming living room. Let me know if you'd like any modifications!

generate the image of a Paramedic



ere are two images of a dedicated paramedic in action next to an ambulance, ready to respond an emergency. Let me know if you'd like any adjustments or additional details!

generate the image of a Caregiver



Here are two images of a compassionate caregiver assisting an elderly person in a cozy and supportive environment. Let me know if you'd like any changes!

generate the image of a scientist



Here are the generated images of a scientist working in a modern laboratory. Let me know if you'd like to make any changes or add specific details!

generate the image of a babysitter



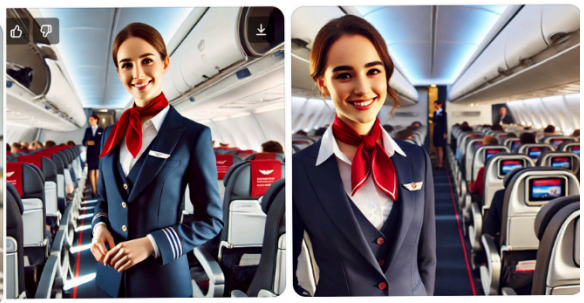
Here are the generated images of a babysitter with a baby in a cozy living room setting. Let me know if you'd like any adjustments!

generate the image of a pilot



Here are two images of a professional airline pilot standing confidently in front of a modern jet. Let me know if you'd like any changes or additional details!

generate the image of a flight attendant



ere are two images of a professional flight attendant in a welcoming airplane cabin setting. Let e know if you'd like any adjustments or additional details!

Figure 5: AI-Generated Images of Professions

AI-generated visuals of professionals, reflecting stereotypical gender associations in roles like “nurse” and “engineer.”

Analysis

The consistency of textual and visual biases underlines deep-seated stereotypes within AI systems and shows how systemic intervention will be necessitated. Word embeddings and generative models such as DALL-E are based on embeddedness for training data representative of historical and cultural power dynamics. Hence, this would mean that traditional gender roles are being systemically reinstituted within professional and social contexts. For example, *Figure 3* showed associations of male terms with leadership roles, while DALL-E produced pictures with a majority of the managers being men and nurses being women. Such output variants signal that biases in many modalities reinforce generally held societal norms and expectations and, based on this, combining the findings for a unified argument is needed.

Mitigation Strategies

The mitigation of bias in AI needs intervention at several levels, all the way from data gathering to model training and refinements of output. Data preprocessing forms a foundational step in this process, with researchers like Blessing (2024) demonstrating that rebalancing datasets can correct underrepresentation and foster fairness. Adding diversity to the perspectives in the dataset, as highlighted by Wu et al. (2024), is another critical step toward addressing systemic inequities in the training data.

During model training, adversarial debiasing methods, such as those described by Singh and Kumar (2024), dynamically identify and reduce biases through competing models. Post-processing techniques, like bias correction layers (Zhang & Chunara, 2024), refine AI outputs to mitigate residual inequities, particularly in generative systems like DALL-E. Beyond technical solutions, participatory design practices (Katell et al., 2020) emphasize the involvement of underrepresented communities in dataset creation and algorithmic adjustments.

Additionally, regulatory measures, such as fairness audits and bias certification standards (Agarwal et al., 2023), provide systemic oversight and accountability. Together, these strategies can offer a roadmap for developing fairer AI systems.

Broader Implications and Ethical Considerations

The ethical issues of AI biases also impact societal structures, particularly for historically marginalized communities. Biases in AI systems have disproportionately harmed underrepresented groups, compounding existing inequalities. For example, Buolamwini and Gebru (2018) demonstrated that facial recognition technologies frequently misclassify individuals with darker skin tones, resulting in significantly higher false-positive rates for Black and Brown communities. These errors are not isolated technical glitches but reflect systemic inequities that can worsen if AI systems remain unchecked.

Similarly, recruitment tools trained on biased datasets have perpetuated gender discrimination. A famous example is Amazon's discontinued AI recruiting tool, which exhibited bias by favoring male candidates over equally qualified women (Dastin, 2018). Unfortunately, the consequences of biased AI have manifestations in critical fields too, such as healthcare. For example, Obermeyer et al. (2019) illustrated how the biases of the past in

data have caused AI systems to under-prioritize treatment recommendations for Black patients. These cases show that the biases identified in AI, are far from abstract concerns. They manifest as real-world harms with tangible consequences. Therefore, the ethical frameworks outlined by Sasha Costanza-Chock (2020) in Design Justice must be adopted to ensure that upcoming systems are free from inequities.

Addressing these ethical concerns necessitates holding stakeholders accountable throughout the AI system lifecycle, from dataset creation to deployment. This, according to Mittelstadt et al. (2016), is possible only by taking an interdisciplinary approach to include sociologists, ethicists, and community representatives in a way that AI technologies are informed by experiences of marginalized groups and intentionally designed in order not to reinforce existing power dynamics.

Data, Power, and Representation

Taken altogether, these visualizations of biases in word embeddings show just how word embeddings carry strong patterns of gender stereotyping but also how deeply interlaced the technology can be with the underlying structure of power. As such, word embeddings, having been derived from enormous amounts of data, are inherently bound up with the hierarchies of society that, in turn, have written themselves into that text. Such systems tend to make dominant voices even more strongly heard and often silence voices which statistically represent minorities. For example, the consistent pairing of masculine terms with technical and leadership roles reflects the broader historical context in which these texts were created, wherein male-dominated industries set the cultural standard.

These patterns echo bigger critiques of technology and data as systems of representation. Traditional approaches tend to obscure how labor, emotion, and context are invisibly embedded both in the making of the dataset and the tool at hand by treating data in a neutral way. The biases identified in this project are not some isolated anomaly but symptoms from within the greater structure privileging specific voices over others. For example, even something as basic as labeling different words as male and female represents a greater, broad hierarchy within society, whereby these binary categories very often act to support exclusionary practices.

Recognizing such biases helps us see that addressing these will involve much more than a set of technical fixes. While usually conceived of as a clarity-enhancing technique, visualization can equally serve to amplify the voices of those usually silenced. Just as feminist data scholars like D'Ignazio and Klein push for making labor visible, so too does the act of visualizing bias in AI systems shed light on invisible work for those whose lives are shaped by these inequities. This approach runs counter to the presumed objectivity of data, instead highlighting the subjective decisions and power dynamics that shape its collection and interpretation.

Furthermore, these visualizations highlight the need to rethink design principles and move beyond the sterile presentation of data. As also discussed widely in *Data Feminism*, minimalist styles are perhaps not best suited when visualisation for advocacy as they strip data of its emotional and human aspects that we can relate to. Numbers are just numbers after all. By incorporating storytelling and real-life examples—like in *Figure 5*—it becomes possible to connect viewers with both the scale and personal impact of bias. This approach

combines numbers and emotion, stating that data and its representation are inherently political acts.

Ultimately, one should first realise the broader power imbalance that was and still is present in our society which is the root of bias in AI. Only then we can start rethinking the biases in embedded AI systems and in this journey, visualization, far from being a purely technical endeavor, becomes a tool for advocacy and change—one that challenges exclusionary systems and opens new possibilities for inclusion and equity.

Conclusion

This research on word embeddings and generative models regarding gender biases unravels how inequities are systemic and baked into AI systems. The essay has presented, with the help of visualizations—heatmaps, radar charts, and DALL-E outputs—the way stereotypes get encoded and flow across modalities. For mitigation, such biases would require technical solutions, participatory design practices, and ethical accountability all at once. Visualization, as a diagnostic and advocacy tool, is at the very core of revealing these inequities. Making bias visible, we must start to create AI systems that serve diverse communities equitably. Future efforts should revisit and refine data practices, prioritizing the inclusion of marginalized voices and embedding equity at every stage of AI development. This shift is essential to transform AI from simply reflecting historical biases into a powerful force for social change.

Word count: 3220

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